

NetPulse

ISP Management System

Installation Guide

Ubuntu 22.04 LTS / 24.04 LTS

One command. Browser wizard. Done.

github.com/YOUR/REPO

Save this guide — update YOUR/REPO placeholder before publishing

QUICK INSTALL (one command)

Run this on a fresh Ubuntu 22.04 or 24.04 server as root:

```
curl -fsSL https://raw.githubusercontent.com/YOUR/REPO/main/deploy/install.sh | sudo bash
```

The script takes 5–8 minutes, then prints your server URL. Open it in a browser and the Setup Wizard launches automatically.

WHAT THE SCRIPT DOES — 9 STEPS

Step	What happens
[1/9] Pre-flight	Checks: Ubuntu 20.04+, RAM "e 1 GB, Disk "e 5 GB, internet connectivity
[2/9] Packages	Installs: git nginx postgresql openssl ufw curl
[3/9] Node.js 24	Via NodeSource apt repo. Skips if Node 24+ already present
[4/9] PostgreSQL	Creates "netpulse" database + user. Auto-generates a secure password
[5/9] Code	Clones repo to /opt/netpulse. Detects existing install !" upgrade mode
[6/9] .env	Writes config with auto-generated SESSION_SECRET & DATABASE_URL. Clerk keys are placeholders (set after install)
[7/9] Build	pnpm install !" build libs !" build API server !" build frontend
[8/9] PM2	Starts app as "netpulse" process, enables systemd auto-start on reboot
[9/9] nginx + UFW	Reverse proxy on port 80. Opens: 22 (SSH), 80 (HTTP), 443 (HTTPS), 1194 (VPN), 1812-1813 (RADIUS) Ends with a health check (curl /api/healthz) and a summary box with your server URL and next steps.

AFTER INSTALL — FIRST RUN WIZARD

1. Open http://YOUR_SERVER_IP in your browser
2. Sign in with your account (set up Clerk first — see next section)

3. The Setup Wizard launches automatically on first login
4. Fill in: Company info !' Timezone & currency !' Billing defaults
5. Click "Launch NetPulse" !' you are on the dashboard
6. The wizard never shows again once completed

SETTING UP CLERK (required for login)

Clerk handles all user authentication. After install:

1. Go to <https://dashboard.clerk.com>
2. Create a Production application (not Development)
3. Under Domains, add your server domain or IP address
4. Copy the Publishable Key and Secret Key
5. Edit `/opt/netpulse/.env` and set the three keys shown below
6. Run `update.sh` to rebuild the frontend with the new keys

```
# Edit this file: /opt/netpulse/.env

CLERK_PUBLISHABLE_KEY=pk_live_xxxxx
CLERK_SECRET_KEY=sk_live_xxxxx
VITE_CLERK_PUBLISHABLE_KEY=pk_live_xxxxx

# Then rebuild:
sudo bash /opt/netpulse/deploy/update.sh
```

HTTPS WITH LET'S ENCRYPT (recommended)

After pointing your domain's DNS A-record to this server:

```
apt install -y certbot python3-certbot-nginx
certbot --nginx -d yourdomain.com
```

Certbot automatically renews the certificate every 90 days. No manual action needed.

UPDATING TO A NEW VERSION

```
sudo bash /opt/netpulse/deploy/update.sh
```

This command: pulls latest code !' installs deps !' builds libs + API + frontend !' runs DB migrations !' restarts PM2 !' health check.

USEFUL COMMANDS

```
pm2 status                # Check app process status
pm2 logs netpulse          # Live application logs
pm2 restart netpulse       # Restart the app
pm2 stop netpulse          # Stop the app

cat /var/log/netpulse/install.log # Full install log
cat /var/log/netpulse/update.log  # Full update log
tail -f /var/log/nginx/netpulse-error.log # nginx errors

sudo -u postgres psql netpulse  # Connect to the database
```

FILE LOCATIONS

Path	What's there
/opt/netpulse/	Application code
/opt/netpulse/.env	Environment config (Clerk keys, DB URL, secrets)
/etc/nginx/sites-available/netpulse	nginx reverse proxy config
/var/log/netpulse/	Install log + update log
/var/log/nginx/netpulse-*.log	Web server access + error logs

MINIMUM SERVER REQUIREMENTS

Resource	Minimum	Recommended
CPU	1 vCPU	2 vCPU
RAM	1 GB	2 GB
Disk	10 GB	20 GB SSD
OS	Ubuntu 22.04 LTS	Ubuntu 24.04 LTS
Ports	22, 80, 443	+ 1194 (VPN), 1812–1813 (RADIUS)

TROUBLESHOOTING

App not responding after install

```
pm2 logs netpulse      # look for startup errors
pm2 restart netpulse
```

Database connection error

```
sudo systemctl status postgresql
cat /opt/netpulse/.env | grep DATABASE_URL
```

nginx 502 Bad Gateway

```
pm2 status              # is netpulse running?
curl localhost:8080/api/healthz  # test API directly
```

Login page blank / Clerk error

- CLERK_PUBLISHABLE_KEY in .env must be pk_live_ (not pk_test_)
- Ensure your domain is added in Clerk dashboard !' Domains
- Run update.sh after changing keys (frontend must be rebuilt with new key)

